



Mineral loads on the Garren and lower Wye

A report prepared on behalf of Wye Salmon Association (WSA)

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Introduction

Wye Salmon Association (WSA) citizen scientist, Peter Hunt, has been monitoring the Garren sub catchment for several years. The Garren has a particular issue with turbidity, even in low summer flows, so Peter was the first volunteer to use the WSA turbidity meter and has been taking regular readings since June 2023.

Such turbidity data naturally begs the question of what quantity of suspended solids is being carried by the river. Turbidity is reported in NTU (nephelometric turbidity units), which is a measure of the amount of light being scattered as a probe beam passes through the sample. Unfortunately, this does not have a fixed relationship with the mass of suspended solids (say as mg/l). Suspended particles can have a wide range of densities, sizes and optical properties, all of which influence scattering and, therefore, turbidity. So, although turbidity meters are calibrated to a defined density of a specific type of particle (formazin) in water, for any other particle type (mineral particles or phytoplankton, say) the relationship is unknown. It will vary by location, flowrate, temperature, time of year, etc.

If we cannot infer a relationship from turbidity to mass of suspended solids, then if we wish to know this, we will need to measure it directly. What follows is a brief description of a piece of work undertaken by Peter to measure the mass of suspended solids under different winter flow conditions on the lower Garren at Packsaddle Bridge, followed by similar measurements from the lower Wye at Redbrook.

Before describing the work, it is useful to understand what scale of numbers might be expected. A recent [press article](#) relating to chinook salmon in the Klamath River in Northern California suggests that suspended solids above 200 mg/l can be detrimental to salmon egg and fry survival. Data is presented showing that peak values in winter can exceed 1000 mg/l on the Klamath, with general levels in the region of 100-200 mg/l.

Method

After performing normal water quality checks, including turbidity, Peter collected a sample of water to take home. One litre of this sample was then subjected to a sequential filtration process in order to determine the mass of suspended solids. A first stage filtration was performed using paper filters, with ratings of 5-13µm, 4-7µm and 1-2µm, using a commercial vacuum assisted filtration system. Filter papers were weighed before use and re-weighed after filtration and drying off, to determine the mass of filtrate. A second stage filtration was then performed using 0.45µm PES membrane cartridge filters (25mm) attached to a 100ml syringe. These were similarly weighed before and after use to determine the mass of filtrate.

Weighing was carried out on a classic chemical balance (which Peter happened to have in his attic). Although capable of mg resolution, this balance is a little past its best and has an estimated reproducibility in the region of 10mg.

Results

Garren at Packsaddle Bridge

We begin the story on the lower Garren immediately below the EA gauging station at Marstow Mill, on Monday 22nd of January. The river was up and coloured after heavy rain, with Marstow Mill recording a 24-hour mean discharge (volumetric flowrate) of 3.85 m³/s – getting on for 4 times the annual average value. Peter recorded a turbidity value of 247 NTU, the highest seen so far on this river. Results from the filtration procedure were as follow in Table 1:

	Filtrate mass (mg/l)	Turbidity (NTU)
Unfiltered		247
First stage filtration down to 1-2µm	195	50
Second stage filtration to 0.45µm	2,190	2
Total	2,485	

Table 1: filtration results for the Garren on 22-January-24

Returning one week later, on 29th of January, Marstow Mill recorded a 24-hour mean discharge of 1.89m³/s and falling. Peter recorded a turbidity of 18 NTU. Filtration results were as follow in Table 2:

	Filtrate mass (mg/l)	Turbidity (NTU)
Unfiltered		18
First stage filtration down to 1-2µm	115	5
Second stage filtration to 0.45µm	1,190	2
Total	1,305	

Table 2: filtration results for the Garren on 29-January-24

Two weeks later, 12th of February, found the Garren high again, this time with a 24-hour mean discharge of 3.05 m³/s. On this occasion, Peter had missed the peak, the river having fined down to a turbidity of 40 NTU. Filtration results were as shown in Figure 3 below:

	Filtrate mass (mg/l)	Turbidity (NTU)
Unfiltered		40
First stage filtration down to 1-2µm	65	
Second stage filtration to 0.45µm	283	
Total	348	

Table 3: filtration results for the Garren on 12-February-24

Wye at Redbrook

Although not a regular monitoring location, the 9th of February found Peter at Redbrook on the lower Wye. The river was high and coloured, with the [Redbrook Gauge](#) showing 4.7m. Data from the [National River Flow Archive](#) tells us that this height corresponds to a discharge of around 390m³/s. Peter recorded a turbidity of 141 NTU and took a sample of water. Filtration results are shown in Figure 4 below. On this occasion a third stage filtration using a 0.22µm PES cartridge filter was performed.

	Filtrate mass (mg/l)	Turbidity (NTU)
Unfiltered		141
First stage filtration down to 1-2µm	345	41
Second stage filtration to 0.45µm	767	13
Third stage filtration to 0.22µm	25	2
Total	1,137	

Table 4: filtration results for the lower Wye on 9-February-24

Discussion

The first observation is that the total suspended solids load in both the Garren Brook and the lower River Wye, during spate conditions, far exceed the 'danger level' for salmon egg & fry survival quoted above from the US work. Indeed, our result from the lower Wye is very similar to the data presented for the Klamath River. Perhaps unsurprisingly, the Garren Brook which directly drains a lot of arable

farmland appears to have substantially higher peak loads than the lower Wye (2,485 mg/l cf. 1,137 mg/l).

Although our first stage filtration, using paper filters rather than membrane types, is not ideal for this purpose, we can tentatively look at the size distribution of suspended solids as summarised in Table 5 below:

	Fraction of particles < 1-2µm
Garren 22-Jan	88%
Garren 29-Jan	91%
Garren 12-Feb	81%
Lower Wye 9-Feb	70%

Table 5: fraction of suspended solids captured on 0.45µm filter
(lower Wye result also includes 2% of 0.22µm filtrate)

This shows that the major fraction of the particle load, in both locations, is micro-meter scale. This is to be expected because the ability of a particle to remain in the water column is driven by the balance between gravity and viscous drag. It can be shown that this scales inversely with particle size, meaning that larger particles tend to drop out of the flow more readily (clogging up the gravel), whereas fine particles remain in suspension longer and tend to deposit in the estuary. The lower Wye appears to carry a slightly higher fraction of larger particles, which may reflect the energetic flow in this section of the river, or perhaps geological differences from the Monnow sub catchment (the Monnow confluence being 3km upstream).

Finally, we can calculate the total load of suspended solids being carried by the two rivers, this being the product of the filtrate mass per litre (from Tables 1-4) and the corresponding river discharge. This is shown in Table 6 below:

	Total mass flow (t/day)
Garren 22-Jan	827
Garren 29-Jan	213
Garren 12-Feb	92
Lower Wye 9-Feb	38,300

Table 6: total loads of suspended solids

Looking first at the Garren on 22nd January, we calculate that this small brook, draining 92 km² of Herefordshire farmland, dumped 827 tonnes of agricultural soil into the river Wye over this 24-hour period. The Wye at Redbrook, draining approximately 4,000km², similarly carried 38,300 tonnes of soil over the 24 hours of 9th February. Both of these numbers correspond to an average soil depth loss in the order of one hundredth of a millimetre per day. So perhaps not significant agronomically, but arguably disastrous for the health of the gravel in the river Wye.

Conclusion

This work represents a quick dip into the measurement of suspended solids in river water. Within the limits of our experimental method, we have clearly demonstrated the scale of the problem, albeit at only two locations. Any further work might benefit from a switch to membrane filters for the first stage filtration and use of a higher precision balance. Although the reproducibility of the current balance is perfectly adequate for spate conditions (the primary focus of this report), any future application to lower flow conditions (and lower filtrate masses) would find this a limitation.

The downstream transport of mineral loads by rivers is a natural process. However, the high level of colour experienced on the Wye and many other rivers today, during spate conditions, is not natural - it is a consequence of agricultural practices. The resulting excessive mineral loads settle out in the river, clogging the gravel to the detriment of all organisms making their home there. We have referred above to the potential impact on salmon reproduction, but other gravel spawning fish and the invertebrates at the base of the aquatic food chain all require clean gravel with water flowing freely through its interstices. It is no surprise that this problem has been of concern for many years now, particularly in the context of the river Wye.

Once again, we find unintended consequences arising from modern, intensive agriculture. We have shown above that the amount of soil loss involved is unlikely to be considered agronomically significant, nevertheless it can cause incalculable harm to the river ecosystem. Parallels with the phosphate situation are inescapable.

A final thought. If I hired a fleet of large tipper trucks and dumped 40 loads of topsoil into the river Wye at Whitchurch, I'd probably end up in jail. But that is pretty much what the farmers of the Garren sub catchment did on 22nd January. And they do it again, every time it rains. And it's happening all over the Wye catchment.

Acknowledgements

I would like to thank long-serving WSA citizen scientist, Peter Hunt for his sterling work on this project. Not only did he develop the concept, but he singlehandedly pulled together the necessary equipment and developed the methodology. Well done.



Peter in his home 'laboratory'. The chemical balance is visible to the right, with the filtration equipment to the rear.